RVCapital — Internal Market Data Repository (IMDR)

Global Blueprint Document

Version: 3.0 **Date:** March 2026 **Owner:** Arjun | Rates & FX Trading

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1. Overview

The IMDR is a centralised internal database consolidating all market data, reference data, and research used by the Rates, Credit, FX, and Macro trading desk. It serves as the single source of truth for pricing, curve construction, trade idea generation, regime detection, and research archival.

Built on **MS SQL Server**. A pristine master dataset is maintained and kept separate — all desk users work off a read-only mirror only.

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2. Design Principles

- **Single DB, multiple schemas** — all asset classes share dimensions, no cross-DB joins
- **Star schema** — fact tables store time series measurements; dimension tables store reference data
- **Surrogate PKs** — all fact tables use `INT IDENTITY(1,1)` as primary key
- **Audit trail** — all tables carry `created_at DATETIME2` defaulting to `SYSUTCDATETIME()`
- **Naming convention** — `fact_` prefix for time series, `dim_` prefix for reference tables
- **Pristine dataset** — master copy is read-only; desk works off a mirrored copy only
- **Bloomberg is non-negotiable** — primary pricing source, never changed without Harmeet's approval
- **Citi Velocity is supplementary only** — carries disclaimers, not closing marks, not for valuations
- **FRED first** — for any public macro series available on both FRED and Bloomberg, use FRED

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3. Database Structure

```
RVCapital_IMDR
│
├── dbo.*          → shared dimensions
├── rates.*        → OIS, govts, swaps, futures, RFRs, vol
├── fx.*           → spot, NDF, forwards, vol surface, fixings
├── credit.*       → CDS, bond spreads, ASW, EM sovereign
├── equities.*     → index futures, vol surfaces, VIX, MOVE
├── commodities.*  → energy, metals, forwards, vol
├── macro.*        → economic data, national statistics, surveys
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├── calendar.*      → events, auctions, holidays, IMM dates
├── positioning.*   → COT, TFF, EPFR, TIC, ETF flows, dealer positions
├── funding.*       → repo, RFR fixings, liquidity, Fed/ECB balance sheets
├── regime.*        → indicators, classification, signals, policy stance
└── research.*      → PDFs, file paths, AI summaries, document tags
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4. Schema Detail

4.1 `dbo` — Shared Dimensions

Goal: Provide a single set of reference lookup tables shared across all schemas. Eliminates duplication of categorical data across fact tables and ensures consistency.

Why it exists: Without shared dimensions every schema would store its own version of currency names, tenor definitions, and instrument types. Joins would break and classifications would drift. The `dbo` schema is the dictionary that the rest of the DB speaks.

Table	Goal	Requirements	Serves
`dim_currency`	Map CCY codes to full names, regions, deliverable status	CCY code (CHAR 3), name, region, deliverable flag (Y/N), onshore/offshore flag	All schemas
`dim_tenor`	Standardise tenor codes across all instruments	Tenor code, approximate days, IMM flag, liquid flag by market	rates, fx, credit, funding
`dim_instrument`	Define all instrument types used across the DB	Instrument code, asset class, description	All schemas
`dim_index`	Reference for all RFR and IBOR indices	Index code, currency, administrator, RFR flag, replacement index	rates, funding
`dim_quote_type`	Define quoting conventions	PAR, ZERO, FWD, ASW, OAS, Z-SPREAD	rates, credit
`dim_country`	Country reference including sovereign metadata	Country code, name, region, sovereign rating, central bank name	All schemas
`dim_bond`	Master bond reference	ISIN, issuer, coupon, maturity, currency, on-the-run flag, index membership	rates, credit
`dim_holiday`	Holiday calendar by country	Country, date, holiday name — affects pricing, fixing, and correlation calcs	calendar, rates, fx

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4.2 `rates` — Rates Schema

Goal: Store the complete interest rate universe — OIS/IRS curves, government bonds, swap spreads, volatility, futures, and RFR fixings. This is the core pricing schema for the rates book.

Why it exists: The rates desk needs a full term structure across multiple currencies and indices at multiple points in the day. Curve construction, DV01 calculation, RV analysis, and regime detection all originate here. Bloomberg is the non-negotiable pricing source approved by Harmeet.

Source rules: Bloomberg primary for all pricing. Citi Velocity supplementary for vol surfaces, historical yields, and swaptions. Last 2 days of Citi Velocity data appended manually from Bloomberg. FRED for public macro rate series.

Table	Goal	Requirements	Serves	Source
fact_ois	Store OIS/IRS par, zero, and fwd curves across all live indices and tenors	Multiple daily snapshots. Full tenor set O/N to 50Y. New curves added dynamically (e.g. onshore India). CCY, index, quote_type, maturity, value	Curve construction, DV01, RV, regime	Bloomberg
fact_fwd_curve	Store forward rate curves by tenor and currency	Fwd rate, tenor start, tenor end, CCY, index	Forward pricing, carry analysis	Bloomberg
fact_fwd_starting_swaps	Store forward starting swap rates	1y1y, 2y1y, 5y5y forward starting rates by CCY	Meeting-dated analysis, CB pricing	Bloomberg
fact_swapspread	Swap rate minus govt bond yield by tenor	Swap rate, bond yield, spread, tenor, CCY	RV between rates and credit, funding premium	Bloomberg
fact_butterfly	Curve and butterfly spread time series	2s5s10s, 2s10s, 5s30s spreads, DV01-weighted and unweighted versions	Curve RV trades, steepener/flattener	Bloomberg / UBS
fact_real_yield	Real yield curves from inflation-linked swaps	Nominal yield, breakeven inflation, real yield by tenor, CCY	Inflation regime, real rate anchor	Bloomberg
fact_xccy_basis	Cross currency basis swap curves full tenor	Basis spread, tenor, CCY pair, collateral type	CCY funding cost, xccy trades	Bloomberg / Citi Velocity
fact_fra_ois	FRA-OIS spread time series	FRA rate, OIS rate, spread, tenor, CCY — funding stress indicator	Funding stress signal, regime input	Bloomberg
fact_basis_swap	3s6s and 1s3s OIS basis	Basis spread by tenor — US, AUD, NZD	Tenor basis trades, curve calibration	Citi Velocity
fact_swaption_vol	Swaption vol surface by expiry x tail	Vol, expiry, tail, CCY, strike offset, normal/lognormal flag	Options pricing, vol RV	Citi Velocity
fact_swaption_straddle	Swaption straddle prices and implied	Straddle price, implied vol, expiry, tail, CCY	Volatility trading, event vol pricing	Citi Velocity

Table	Goal	Requirements	Serves	Source
	vols			
fact_vol_surface	Full rates vol surface	ATM, 25d calls, puts, skew by CCY and tenor — USD, EUR, AUD, GBP, JPY, KRW, NZD	Vol surface analysis, skew trades	Citi Velocity
fact_vol_realised_implied	Realised vs implied vol spread	Realised vol, implied vol, spread, rolling windows, CCY	Vol regime, carry in vol space	Citi Velocity
fact_receiver_payer_skew	Receiver vs payer skew across tenors	Skew, tenor, CCY, expiry	Directional vol bias, market positioning	Citi Velocity
fact_rfr_fixing	Daily RFR fixings	SOFR, SONIA, €STR, TONAR, AONIA, SORA — date, fixing value, CCY, index	OIS curve calibration, compounded rates	Bloomberg / FRED
fact_bond_futures	Bond futures pricing and basis analysis	Futures price, CTD bond, net basis, gross basis, implied repo, delivery date	Futures basis trades, CTD switches	Bloomberg / Repo Emails
fact_bond_futures_roll	Futures roll richness/cheapness	Roll spread, front/back contract, CCY, roll date	Roll timing, calendar spread trades	Bloomberg
fact_govtbond	Government bond prices and analytics	Price, yield, ASW, z-spread, DV01, on/off-the-run spread, bid-ask spread by ISIN	Bond RV, duration positioning, ASW trades	Bloomberg
fact_govtbond_historical	Historical bond yields including matured bonds	5-10yr history, ISIN, yield, date — includes bonds no longer live	Back-testing, historical RV analysis	Citi Velocity
fact_asw	Asset swap spreads by bond	ASW spread, bond ISIN, tenor, CCY, swap type	ASW trades, credit-rates RV	Bloomberg
fact_repo	Repo rates by bond — GC vs specials	Repo rate, bond ISIN, collateral type, GC/special flag, CCY — JGB and US	BOJ market stress, implied financing	Barclays / PAML Emails (AI Parsed)

Table	Goal	Requirements	Serves	Source
fact_strips	STRIPS zero coupon curve data	Strip price, yield, maturity, CCY — zero coupon curve construction	Zero curve construction, convexity	Bloomberg
fact_imm_dates	IMM dates and meeting-dated OIS	IMM date, meeting date, days to meeting, OIS rate priced for meeting	CB meeting pricing, short end analysis	Bloomberg
fact_term_premium	Term premium estimates	ACM model, Kim-Wright model estimates by tenor — from NY Fed	Term premium decomposition, regime	FRED / NY Fed
fact_yield_curve_slope	Yield curve slope history	10y2y, 10y3m slopes by CCY — inversion/steepening classification	Regime signal, recession indicator	FRED
fact_fin_conditions	Financial conditions indices	Chicago Fed NFCI, Goldman FCI — weekly	Broad macro tightening/easing signal	FRED

4.3 fx — FX Schema

Goal: Store FX spot, NDF, forward, and vol data across deliverable and non-deliverable pairs. Template for how rates data should eventually be structured.

Why it exists: FX is directly linked to rate differentials through covered interest parity. Forward points embed the rate differential. NDF rates for restricted currencies (CNY, INR, KRW, TWD, IDR) are the only tradable offshore instrument. Vol surface feeds into cross-asset regime detection. BidFX is the primary source, already cleaned and validated by Rajesh and Yash, migrating to Alliance with read-only access.

Table	Goal	Requirements	Serves	Source
fact_ohlc	FX spot, NDF, forward OHLC bars	Hourly bars, open/high/low/close, bid/ask/mid, n_ticks, symbol, series, tenor	FX RV, entry/exit, liquidity monitoring	BidFX API
fact_fwd_points	FX forward points by tenor	Forward points, tenor, CCY pair — directly linked to rate differentials	CIP analysis, rates-FX RV	BidFX API
fact_fwd_curve	Full deliverable forward curve	Forward rate, tenor, CCY pair	Forward pricing, carry trades	BidFX API
fact_ndf_rates	NDF rates for restricted CCYs	NDF rate, tenor, CCY — CNY, INR, KRW, TWD, IDR	EM FX positioning, offshore pricing	BidFX API
fact_fixing	Daily FX fixings	WMR, MAS, RBI fixings by CCY and date	Fixing risk, benchmark valuation	Bloomberg
fact_vol_surface	FX implied vol surface	ATM, 25d RR, 25d BF by tenor and CCY pair	Options pricing, vol regime, hedging cost	Bloomberg / Citi Velocity

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4.4 credit — Credit Schema

Goal: Store credit spread data across indices, single names, and EM sovereign CDS. Monitor credit-rates correlation as a regime signal.

Why it exists: Credit spreads widen ahead of risk-off moves in rates. EM sovereign CDS is a direct signal for EM rates trades. CDX and iTraxx are liquid macro instruments for risk hedging. Credit-rates correlation is a core regime input — when they decouple something is breaking.

Table	Goal	Requirements	Serves	Source
fact_cds_index	CDS index levels	CDX IG, CDX HY, iTraxx Main, iTraxx Crossover — series, tenor, spread, price	Macro hedging, risk-off signal	Citi Velocity
fact_cds_singlename	Single name CDS spreads	Spread, tenor, entity name, CCY, tier	Credit RV, name-specific risk	Citi Velocity
fact_cds_em_sovereign	EM sovereign CDS by country	5Y benchmark spread, country, CCY, date	EM rates trades, sovereign risk	Citi Velocity
fact_bond_spread	Corporate bond spreads by rating	OAS, z-spread, spread by rating bucket — AA, AAA, BBB — sector, CCY	Credit regime, spread compression trades	Citi Velocity
fact_asw	Asset swap spreads by bond	ASW spread, ISIN, CCY, tenor	Govt bond richness/cheapness vs swaps	Bloomberg
fact_credit_corr	Credit-rates rolling correlation	Rolling correlation between spread changes and rate moves — regime input	Regime detection, cross-asset signals	Derived

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4.5 equities — Equities Schema

Goal: Store equity index futures, vol surfaces, and cross-asset signals that matter for a macro rates fund. Equities here are inputs to regime detection, not an equity book.

Why it exists: Equity vol (VIX) and rates vol (MOVE) together define the risk environment. Equity-bond correlation is the single most important regime variable — when it flips from negative to positive the entire rates hedge framework changes. Sector rotation signals growth vs inflation regimes.

Table	Goal	Requirements	Serves	Source
fact_index_futures	Equity index futures prices	SPX, SX5E, KOSPI, Nikkei, ASX — price, volume, open interest, date	Risk-on/off signal, regime input	Bloomberg
fact_equity_vol	Equity vol surfaces and skew	SPX, SX5E, KOSPI, Nikkei — ATM vol, 25d skew, put/call spread by tenor	Cross-asset vol regime, tail risk	Citi Velocity
fact_vix	VIX term structure and futures	VIX spot, VIX futures by expiry, term structure slope, contango/backwardation	Risk-off signal, vol regime	Bloomberg
fact_move	MOVE index time series	MOVE index level, 1m, 3m realised — rates vol equivalent of VIX, core regime variable	Rates vol regime, risk-off signal	Bloomberg
fact_equity_bond_corr	Rolling equity-bond correlation	30d, 60d, 90d rolling correlation — positive = inflation regime, negative = growth regime	Regime classification anchor	Derived
fact_sector_rotation	Equity sector rotation	Cyclicals vs defensives relative performance, rolling ratio — growth proxy	Macro regime signal, growth vs recession	Bloomberg

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4.6 commodities — Commodities Schema

Goal: Store energy and metals prices as macro regime inputs. Oil drives inflation expectations. Gold is the real rates proxy. Copper is the global growth proxy.

Why it exists: A rates macro fund does not trade commodities directly but cannot ignore them. Oil is the single biggest input into CPI and therefore CB reaction functions. Gold moves inversely to real rates — it is a real-time cross-check on the real yield framework. Copper leads global PMI by several weeks. These are not optional regime inputs.

Table	Goal	Requirements	Serves	Source
fact_energy_spot	WTI and Brent crude spot prices	Daily spot price, CCY, date	CPI forecast, inflation regime, CB reaction	Bloomberg
fact_energy_fwd	Oil forward curves	Forward price by tenor, WTI and Brent, backwardation/contango flag	Supply/demand signal, inflation expectations	Bloomberg
fact_metals_spot	Gold, Copper, Silver spot prices	Daily spot price, CCY, date — Gold = real rate proxy, Copper = growth proxy	Real rates validation, growth regime	Bloomberg
fact_metals_fwd	Metals forward curves	Forward price by tenor — Gold, Copper	Carry in metals, forward real rate signal	Bloomberg
fact_commodity_vol	Commodity implied volatility	ATM vol by tenor — oil and metals	Commodity risk regime, tail risk pricing	Bloomberg

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4.7 macro — Economic Data Schema

Goal: Store the full set of macro economic releases and survey data across countries. This is the fundamental data layer underpinning CB reaction function modelling and regime detection.

Why it exists: Every rates trade has a macro thesis. CPI drives CB decisions which drive the short end. GDP and labour data drive the belly. Surprise indices tell you when the market is being caught offside. Without a clean, structured macro layer the regime detection framework has no inputs.

Table	Goal	Requirements	Serves	Source
fact_cpi	CPI time series by country	Headline, core, core-core, super core (US), tradable vs non-tradable breakdown, MoM and YoY	CB reaction function, inflation regime	FRED / National Stats
fact_pce	PCE deflator — Fed preferred inflation gauge	Headline and core PCE, MoM and YoY	Fed reaction function modelling	FRED
fact_ppi	PPI — pipeline inflation pressure	PPI by country, input/output split, MoM and YoY	Leading inflation signal	FRED / National Stats
fact_inflation_expectations	Market and survey inflation expectations	5y5y breakeven, University of Michigan 1y and 5y, NY Fed survey, TIPS breakeven by tenor	Inflation regime anchor, real rate calc	FRED / Bloomberg
fact_gdp	GDP by country	Quarterly GDP, components — consumption, investment, govt, net exports	Growth regime, CB reaction	FRED / National Stats
fact_trade	Trade balance and current account	Exports, imports, net exports, current account balance by country	FX regime, external balance signal	FRED / National Stats
fact_fiscal	Fiscal balance by country	Budget deficit, debt-to-GDP, fiscal impulse — affects duration supply	Supply/demand for bonds, term premium	National Stats
fact_labour	Labour market data	Unemployment, NFP, JOLTS openings and quits, wage growth, average hourly earnings, participation rate	CB reaction function, late cycle signal	FRED
fact_surveys	Business and consumer surveys	ISM/PMI manufacturing and services by country, retail sales, consumer confidence, UoM sentiment	Leading economic indicators, regime	FRED / Bloomberg
fact_housing	US housing market data	Housing starts, building permits, existing and new home sales	Rate-sensitive sector signal	FRED

Table	Goal	Requirements	Serves	Source
fact_industrial	Industrial activity	Industrial production, durable goods orders, capex in GDP by country	Growth proxy, investment cycle	FRED / National Stats
fact_money_supply	Money supply by country	M1, M2 by country and date	Monetary regime, liquidity signal	FRED / National Stats
fact_bank_lending	Bank lending surveys	Fed Senior Loan Officer survey, ECB BLS, loan growth by country — household, housing, business	Credit tightening signal, recession risk	FRED / ECB
fact_surprise_index	Economic surprise indices	Citi Economic Surprise Index, JPM equivalent by country	Market positioning vs data reality	Citi Velocity / Bloomberg
fact_social_financing	China and Korea credit data	China total social financing, new yuan loans, aggregate financing. Korea social financing	China macro regime, EM rates signal	National Stats (PBoC, BOK)
fact_property	Property prices by major city	Price index by city and country — quarterly or monthly	Rate-sensitive asset, CB concern	National Stats
fact_terms_of_trade	Terms of trade by country	Export price index / import price index ratio	External competitiveness, FX regime	National Stats
fact_ubs_forecasts	UBS macro forecasts	Policy rates, CPI, GDP, exports forecasts by country — monthly/weekly frequency	Consensus vs positioning, trade thesis	UBS Economic Files
fact_policy_rates	Historical CB policy rates	Rate on each meeting date, CB name, country, direction — hiking/cutting/hold	CB reaction function, regime classification	FRED / UBS

4.8 calendar — Events & Calendar Schema

Goal: Track all scheduled and unscheduled events that move rates markets. The calendar layer is as important as the pricing layer on a rates desk.

Why it exists: Rates markets are event-driven. A position that is correctly sized for normal market conditions can be catastrophically wrong into an unhedged CB meeting or auction tail. The calendar schema ensures every event is known, tracked, and linked to relevant data.

Table	Goal	Requirements	Serves	Source
fact_cb_events	Central bank event calendar	Meeting dates, MPC minutes dates, press conference times, Fed speak events, speaker name, CB name	Event risk management, meeting-dated OIS	Bloomberg / CB Websites
fact_cb_balance_sheet	CB balance sheet schedule	QE/QT monthly pace, reinvestment dates, asset class, CB name	Duration supply, term premium	CB Websites
fact_soma_calendar	SOMA and ECB reinvestment calendar	Operation dates, size, maturity of reinvested bonds — ECB APP/PEPP	Supply/demand for specific tenors	NY Fed / ECB
fact_auction_calendar	Govt bond auction calendar	Auction date, maturity, size, new issue / reopening, CCY, bid-to-cover, tail vs WI	Auction risk, duration supply tracking	Bloomberg / DMOs
fact_treasury_refunding	US Treasury refunding	Refunding announcement dates, quarterly borrowing estimate, auction sizes by tenor	Term premium, supply shock risk	US Treasury / FRED
fact_econ_releases	Economic release calendar	Release date, indicator name, country, frequency, prior value, consensus, actual	Event risk, surprise index construction	Bloomberg
fact_debt_ceiling	Debt ceiling and fiscal deadlines	X-date estimates, legislative deadlines, TGA depletion forecast	T-bill supply shock, liquidity risk	US Treasury / Bloomberg
fact_index_rebal	Bond index rebalancing dates	Rebalancing date, index name, estimated flow direction, estimated size	Predictable flow, front-running signal	Bloomberg
fact_options_expiry	Swaption and futures options expiry	Expiry date, instrument type, CCY, tenor, estimated open interest	Gamma exposure, vol into expiry	Bloomberg
fact_imm_roll	IMM dates and futures roll	IMM date, front contract, back contract, roll period, CCY	Futures roll timing, short end calendar	Bloomberg
fact_isda_fixing	ISDA fixing dates	Fixing date, index, CCY — affects swap cash flows	Swap cash flow management	ISDA

Table	Goal	Requirements	Serves	Source
fact_elections	Election calendar	Election date, country, type (general, presidential), current poll data	Political risk, fiscal regime change	News / Bloomberg
dim_holiday	Holiday calendar by country	Country, date, holiday name, market impact flag	Pricing adjustments, settlement dates	Bloomberg / National Sources

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4.9 positioning — Positioning & Flow Schema

Goal: Track who is holding what, in which direction, and at what size across the rates and macro universe. Positioning is the context without which price data alone is incomplete.

Why it exists: Price levels tell you where the market is. Positioning tells you who is vulnerable and where the squeeze risk lies. A 10Y yield at 4.5% means something very different if leveraged funds are max short vs max long. EPFR flows tell you whether retail is buying or selling into a move. TIC tells you whether foreign CBs are reducing UST holdings. These are not optional signals — they are essential context for every trade.

Table	Goal	Requirements	Serves	Source
fact_cftc_cot	CFTC Commitment of Traders	Weekly. Treasury futures (2Y, 5Y, 10Y, 30Y), SOFR futures — non-commercial, commercial, net position, open interest	Leveraged fund positioning, squeeze risk	CFTC
fact_cftc_tff	Traders in Financial Futures report	Weekly. Leveraged funds vs asset managers vs dealers — net position, changes week-on-week	Institutional positioning split	CFTC
fact_dv01_positioning	DV01-weighted futures positioning	Convert TFF net positions to DV01 — comparable across tenors	Duration risk in futures market	Derived from CFTC
fact_epfr_flows	EPFR bond fund flows	Weekly inflows/outflows by region — EM, DM, HY, IG, govt — USD amount and as % of AUM	Retail and institutional flow signal	EPFR
fact_etf_flows	Bond ETF flow data	Daily flows — TLT, IEF, SHY, BND and equivalents — shares created/redeemed, AUM change	Real-time retail positioning signal	EPFR
fact_primary_dealer	Primary dealer positions	Weekly. Net long/short by tenor bucket — Fed weekly H.15 publication	Dealer capacity and warehousing risk	Fed
fact_tic	Treasury International Capital	Monthly. Foreign holdings of USTs by country — total and change	Foreign CB demand, geopolitical flow risk	TIC / US Treasury
fact_fed_custody	Fed custody holdings	Weekly. Securities held in custody for foreign CBs — proxy for foreign CB UST demand	Indirect demand signal	Fed
fact_cb_reserves	Central bank FX reserves	Monthly. BOJ, ECB, PBOC reserve levels, changes, and intervention signals	FX intervention risk, UST demand	National Central Banks
fact_clearing_notional	Swap clearing notional outstanding	Monthly. JSCC, LCH, CME notional by CCY and tenor — open interest in cleared swaps	Market size, systemic risk monitoring	JSCC / LCH / CME

4.10 funding — Funding & Liquidity Schema

Goal: Monitor the plumbing of the financial system — repo, RFR fixings, central bank balance sheets, and liquidity conditions. Funding stress is the earliest warning signal of systemic risk in rates markets.

Why it exists: When funding markets are stressed everything else reprices. FRA-OIS widens. Cross-currency basis blows out. JGB specials spike when BOJ is squeezed. TGA drawdowns inject reserves into the system and compress short rates. The Fed RRP facility tells you how much excess liquidity is parked away from the market. Without this schema the regime framework is blind to the most important early warning signals in rates.

Table	Goal	Requirements	Serves	Source
fact_rfr_fixing	Daily RFR fixings	SOFR, SONIA, €STR, TONAR, AONIA, SORA — daily fix, compounded index	OIS curve calibration, funding cost baseline	Bloomberg / FRED
fact_repo_us	US repo markets	SOFR, tri-party, GCF repo rates — daily, by tenor	USD funding conditions, bill-OIS signal	Bloomberg
fact_repo_jgb	JGB repo and specials	Daily repo and reverse repo rates on JGBs, specials identification, extracted from historical emails via AI	BOJ market stress, short squeeze risk in JGBs	Barclays / PAML Emails (AI Parsed)
fact_fed_rrp	Fed Reverse Repo facility	Daily RRP usage and counterparty count — excess liquidity gauge	Reserves level, short rate floor	FRED
fact_fed_reserves	Bank reserves at the Fed	Weekly bank reserves — leading indicator of QT impact	Liquidity regime, reserves scarcity risk	FRED
fact_tga	Treasury General Account balance	Daily TGA balance — drawdowns inject reserves, rebuilds drain them	Short rate impact, bill supply signal	FRED
fact_fed_balance_sheet	Fed balance sheet composition	Weekly. Treasuries, MBS, reserves, RRP, TGA — full composition	QT pace tracking, duration supply	FRED
fact_ecb_facility	ECB deposit facility and TLTRO	ECB deposit facility usage, LTRO/TLTRO outstanding, PEPP/APP reinvestment pace	EUR funding conditions, ECB QT impact	ECB
fact_bill_ois	T-bill vs OIS spread	T-bill yields across tenors vs OIS — bill-OIS spread signals collateral scarcity or supply shocks	Collateral market conditions, debt ceiling signal	Bloomberg
fact_fra_ois	FRA-OIS spread	FRA rate vs OIS rate by tenor and CCY — funding stress and credit risk in interbank market	Systemic funding stress signal	Bloomberg

4.11 `regime` — Regime Detection Schema

Goal: Classify the current macro regime using ~20-30 global indicators drawn from all other schemas. Every trading strategy on the desk must have a regime filter — a steepener that works in a cutting cycle destroys money in a hiking cycle.

Why it exists: The IMDR is not just a data store. Its end goal is to power a regime detection framework that tells the desk what environment they are operating in before any trade is put on. The regime schema is the brain built on top of the rest of the DB. It should explain approximately 90% of global market moves using a compact set of core variables.

Table	Goal	Requirements	Serves	Source
fact_regime_indicators	Store the ~20-30 core global regime indicators	Daily. Oil, S&P, Nikkei, VIX, MOVE, Fed funds, Asia FX basket, DXY, BBDXY, real 10Y yield, 10y2y slope, xccy basis, credit spreads, equity-bond corr, EM-DM rate differential	Regime classification input	Derived from all schemas
fact_regime_classification	Daily regime output	Regime type, confidence score, date, key indicator contributions — growth, inflation, risk-off, liquidity stress, dollar squeeze	Trade idea filtering, strategy validation	Derived
fact_cb_policy_stance	CB policy stance classification	Per CB per meeting — hiking, pausing, cutting, QE, QT, forward guidance classification	Regime input, rates strategy framing	Derived / Bloomberg
fact_dollar_regime	USD regime classification	DXY, BBDXY levels and trend, strong/weak classification, rolling z-score	EM rates and FX regime input	Derived / Bloomberg
fact_em_dm_differential	EM vs DM rate differential	Rate differential by CCY pair, z-score, trend direction	EM carry regime, xccy basis signal	Derived
fact_equity_bond_regime	Equity-bond correlation regime	30d, 60d, 90d rolling corr, regime classification — positive = inflation, negative = growth	Core hedge framework switch signal	Derived
fact_curve_regime	Yield curve shape regime	Slope level, trend, classification — steepening, flattening, inverted, bear/bull	Strategy regime filter	Derived
fact_real_rate_regime	Real rate level and regime	Real 10Y yield level, z-score, trend, regime classification — deeply negative, neutral, restrictive	Valuation anchor for all rates trades	Derived
dim_regime_type	Regime type definitions	Regime name, description, typical asset class behaviour, historical examples	Reference for regime classification	Static

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4.12 research — Research Schema

Goal: Give every piece of research, trade idea, and reference document a structured home with AI-generated summaries and searchable tags. Eliminate the problem of institutional knowledge living in inboxes and shared drives.

Why it exists: The desk receives research from UBS, Citi, Bloomberg, and internal sources daily. Without a structured layer this knowledge is lost in email threads. The research schema ingests every PDF, generates a summary via AI, and tags it by asset class, currency, tenor, and topic. Over time this becomes the desk's institutional memory and a key input into trade idea generation.

Table	Goal	Requirements	Serves	Source
fact_document	Master document registry	One row per document — title, file path, file name, published date, received date, document type, source	All desk users — searchable research library	All sources
fact_document_summary	AI or manual summary per document	Summary text, summary type (AI/MANUAL), model used, document_id FK, created_at	Quick research retrieval, trade idea generation	AI Pipeline (Claude)
fact_document_tag	Tag each document	Tag type (ASSET_CLASS, CCY, TENOR, TOPIC), tag value, document_id FK — enables multi-dimensional search	Filtered research retrieval by trade context	Derived
dim_document_type	Document type reference	Report, Trade Idea, Central Bank Minutes, Economic Release, Repo Email, Internal Note	Document classification	Static
dim_source	Research source reference	UBS, Citi, Bloomberg, Internal, Fed, MAS, Barclays, PAML, ECB, BOJ	Source attribution, Citi disclaimer flagging	Static

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5. Data Sources Summary

Source	Schemas	Type	Key Rules
Bloomberg	rates, fx, equities, commodities, funding, calendar	Primary pricing	Non-negotiable. Never change without Harmeet's approval
BidFX API	fx	Streaming FX quotes	Hourly OHLC. Cleaned by Rajesh and Yash. Migrating to Alliance read-only

Source	Schemas	Type	Key Rules
Citi Velocity	rates, credit, equities, macro	Supplementary / delayed	Not closing marks. Not for valuations. Last 2 days appended manually from Bloomberg
FRED	rates, macro, funding, positioning	Public macro	Free and reliable — use before Bloomberg for any public series
UBS Economic Files	macro	Internal research	Policy rates, CPI, GDP, exports, forecasts. Monthly/weekly
Barclays / PAML Emails	rates, funding	AI Parsed	JGB repo, specials, implied repo — historical email extraction
CFTC	positioning	Positioning	COT and TFF weekly reports
EPFR	positioning	Flow data	Bond fund and ETF flows weekly
TIC / Fed	positioning, funding	Flow and liquidity	Foreign CB holdings, custody, reserves, primary dealer positions
National Stats Offices	macro, calendar	Government data	Irregular publication schedules — system must scrape and track
JSCC / LCH / CME	positioning	Clearing data	Swap notional outstanding monthly
ECB / BOJ / PBOC	macro, funding, positioning	Central bank data	Balance sheet, facility usage, reserve data
NY Fed	rates	Research data	Term premium models — ACM, Kim-Wright

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6. Data Types

MS SQL Type	Used For
`INT IDENTITY(1,1)`	Surrogate primary key on all fact tables
`DATETIMEOFFSET(0)`	All timestamps — preserves UTC offset
`DECIMAL(18,6)`	Rates, yields, spreads
`DECIMAL(22,8)`	FX prices — handles small (0.69) and large (16,800) values
`CHAR(3)`	ISO currency codes
`VARCHAR(n)`	Categorical fields

MS SQL Type	Used For
`NVARCHAR(MAX)`	Research summary text
`DATE`	Publication and reference dates
`INT`	Counts, foreign keys
`DATETIME2`	Audit timestamps
`BIT`	Boolean flags — deliverable, on-the-run, special

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7. Key Rules

- Bloomberg is the primary pricing source — **never changed without Harmeet's approval**
- Citi Velocity carries disclaimers — **not closing marks, not for valuations**
- Pristine master dataset is kept separate — **desk works off a read-only mirror only**
- Citi Velocity used for supplementary data, back-testing, and gap-filling only
- FRED is used before Bloomberg for any public macro series available on both

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8. Build Plan

Phase 1 — Foundation

- [ ] Create database `RVCapital_IMDR`
- [ ] Create all 12 schemas
- [ ] Build all `dbo` dimension tables
- [ ] Build `rates.fact_ois` and `fx.fact_ohlc` — first two tables already spec'd

Phase 2 — Rates Buildout

- [ ] Build all `rates.*` fact tables
- [ ] Connect Bloomberg — OIS, govts, swapspread, butterfly, futures, vol
- [ ] Connect FRED — RFR fixings, term premium, yield curve slope, fin conditions
- [ ] Set up AI parser for Barclays/PAML repo emails

Phase 3 — FX Buildout

- [ ] Build all `fx.*` fact tables
- [ ] Migrate BidFX API to Alliance read-only
- [ ] Connect Bloomberg for vol surface and fixings

Phase 4 — Credit Buildout

- [ ] Build all `credit.*` fact tables
- [ ] Connect Citi Velocity for CDS and bond spread data

Phase 5 — Macro and Calendar

- [ ] Build all `macro.*` and `calendar.*` tables
- [ ] Set up FRED ingestion pipeline
- [ ] Build national stats scraper with irregular schedule tracker
- [ ] Connect UBS economic files

Phase 6 — Positioning and Funding

- [ ] Build `positioning.*` and `funding.*` tables
- [ ] Set up CFTC, EPFR, TIC feeds
- [ ] Connect Fed, ECB balance sheet and TGA data
- [ ] Connect Barclays/PAML email parser for JGB repo

Phase 7 — Equities and Commodities

- [ ] Build `equities.*` and `commodities.*` tables
- [ ] Connect Bloomberg for index futures and commodities
- [ ] Connect Citi Velocity for equity vol surfaces

Phase 8 — Research Layer

- [ ] Build `research.*` schema
- [ ] Build PDF ingestion pipeline
- [ ] Integrate AI summarisation pipeline (Claude API)
- [ ] Build document tagging logic

Phase 9 — Regime Detection

- [ ] Build `regime.*` schema
- [ ] Define and populate ~20-30 global regime indicators
- [ ] Build classification logic targeting ~90% explanatory power
- [ ] Build dashboard with regime overlay and custom model analysis
- [ ] Every strategy on desk gets a regime filter built in